

SUPPLY CHAIN ANALYSIS

End-to-End Data Analytics Project using SQL & Power BI

Project Documentation

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1. Introduction

Supply chains generate large volumes of operational data every day — from raw material procurement and manufacturing to shipping, inspection, and final sale. When this data is scattered across spreadsheets and disconnected systems, decision-makers struggle to see the bigger picture. This project analyses an end-to-end supply chain dataset covering product manufacturing, supplier performance, inventory, logistics, and profitability, and converts it into a set of interactive Power BI dashboards backed by structured SQL analysis.

The resulting solution allows stakeholders — operations managers, procurement teams, and business leaders — to monitor performance in real time, identify bottlenecks, and make data-driven decisions instead of relying on intuition or delayed manual reports.

2. Project Overview

The **Supply Chain Analysis** project is an analytics case study built around three interconnected product categories — skincare, haircare, and cosmetics — sourced from five suppliers and distributed through four transportation modes (Air, Road, Rail, and Sea). The project follows a complete analytics pipeline: raw data ingestion, cleaning and transformation, relational database design, SQL-based exploratory analysis, and finally an interactive multi-page Power BI dashboard suite covering revenue, inventory, logistics, and profitability.

The dashboards are organised into four focused views — Executive Overview, Inventory & Supplier Analysis, Logistics & Profitability Analysis, and Business Insights & Recommendations — so that different stakeholders can quickly access the metrics most relevant to them.

3. Business Problem

Organisations managing multi-supplier, multi-modal supply chains often face difficulty answering basic but critical questions quickly: Which products drive the most revenue? Which suppliers are reliable versus prone to defects and delays? Which transportation mode is the most cost-efficient for a given product line? Without a consolidated analytical view, these questions require manual cross-referencing of multiple spreadsheets, leading to slow and error-prone decisions.

- Revenue and defect-rate trends are not visible at a glance across product categories.
- Supplier performance (production volume, defect rate, lead time) is not benchmarked centrally.
- Shipping cost and shipping time vary significantly by transportation mode, but this was not being tracked systematically.
- Profitability by product type was not linked back to logistics and manufacturing cost drivers.

4. Objectives

- Design a clean, query-ready relational database from raw supply chain data.
- Use SQL to explore revenue, defect rates, stock levels, and supplier performance.
- Build a Power BI data model connecting products, suppliers, logistics, and sales facts.
- Create DAX measures for KPIs such as total revenue, average defect rate, and profit margin.
- Develop an interactive, multi-page dashboard suite for different business functions.
- Translate dashboard findings into actionable business recommendations.

5. Scope of the Project

The scope of this project covers analysis across three product types (skincare, haircare, cosmetics), five suppliers, and four transportation modes (Road, Air, Rail, Sea). It spans the full analytics lifecycle — from raw data to a decision-ready dashboard — but does not include real-time data streaming, predictive/forecasting models, or integration with a live ERP system. All analysis is based on a static historical snapshot of supply chain records.

6. Tools & Technologies

Category	Tool / Technology	Purpose
Database	MySQL	Data storage and SQL-based analysis
BI & Visualization	Microsoft Power BI Desktop	Data modelling, DAX measures, dashboards
Data Preparation	Power Query (M)	Data cleaning and transformation
Modelling Language	DAX (Data Analysis Expressions)	KPI and measure creation
Spreadsheet Tools	Microsoft Excel / CSV	Initial raw data inspection

7. Dataset Description

The dataset represents supply chain operations for a consumer goods business selling skincare, haircare, and cosmetics products. It captures product-level, supplier-level, and shipment-level detail sufficient to analyse revenue, quality, inventory, and logistics performance.

Field	Description
Product Type	Category of the product — skincare, haircare, or cosmetics
Supplier Name	One of five suppliers (Supplier 1 – Supplier 5)
Revenue Generated	Sales revenue attributed to each product / supplier
Stock Level	Current inventory quantity on hand
Production Volume	Units manufactured, by supplier
Manufacturing Cost	Average cost to produce one unit
Defect Rate	Percentage of defective units
Inspection Result	Pass / Fail / Pending quality check outcome
Transportation Mode	Road, Air, Rail, or Sea
Shipping Cost	Cost incurred to ship products
Shipping Time	Time taken (days) to deliver shipment
Supplier Lead Time	Average lead time per supplier (days)
Profit Margin	Profitability percentage by product type

8. Database Design

The raw data was normalised into a relational schema in MySQL to support efficient querying and to eliminate redundancy. The design separates dimension entities (Product, Supplier, Transportation Mode) from a central fact table capturing transactional/shipment-level measures.

Core Tables

- **dim_product** — product_id, product_type
- **dim_supplier** — supplier_id, supplier_name
- **dim_transport_mode** — mode_id, mode_name
- **fact_supply_chain** — links product, supplier, and transport mode with revenue, stock level, production volume, manufacturing cost, defect rate, inspection result, shipping cost, shipping time, and lead time

This star-schema-style structure keeps dimension attributes (like supplier name or product type) stored once, while the fact table holds the numeric measures used throughout the SQL analysis and the Power BI data model.

9. Data Cleaning & ETL

Before analysis, the raw dataset was cleaned and standardised using Power Query and SQL. Key cleaning steps included:

- Removing duplicate shipment and inventory records.
- Standardising text values (e.g. consistent casing for product type and supplier names).
- Handling missing or null values in defect rate and shipping cost fields.
- Converting currency and percentage fields to correct numeric data types.
- Validating that inspection results only contained the expected categories: Pass, Fail, Pending.
- Creating derived columns such as total logistics cost (shipping cost + manufacturing cost) for later analysis.

10. SQL Analysis

Exploratory analysis was performed directly in MySQL to validate assumptions and pre-answer key business questions before dashboard design. Representative queries included:

Revenue by Product Type

```
SELECT product_type, SUM(revenue_generated) AS total_revenue
FROM fact_supply_chain f JOIN dim_product p ON f.product_id = p.product_id
GROUP BY product_type ORDER BY total_revenue DESC;
```

Average Defect Rate by Supplier

```
SELECT s.supplier_name, AVG(defect_rate) AS avg_defect_rate
FROM fact_supply_chain f JOIN dim_supplier s ON f.supplier_id = s.supplier_id
GROUP BY s.supplier_name ORDER BY avg_defect_rate DESC;
```

Average Shipping Cost by Transportation Mode

```
SELECT t.mode_name, AVG(shipping_cost) AS avg_shipping_cost
FROM fact_supply_chain f JOIN dim_transport_mode t ON f.mode_id = t.mode_id
GROUP BY t.mode_name ORDER BY avg_shipping_cost DESC;
```

These queries confirmed early patterns — skincare leading in revenue, Supplier 1 leading in overall contribution, and Air freight carrying the highest average shipping cost — that were later visualised in Power BI.

11. Power BI Data Model

The cleaned MySQL tables were imported into Power BI and connected in a star schema. The fact table (supply chain shipment records) sits at the centre, with one-to-many relationships to the Product, Supplier, and Transportation Mode dimension tables. This structure keeps the model efficient and allows slicers such as Product Type, Supplier Name, and Transportation Mode to filter every visual consistently across all report pages.

- Relationships set to single-direction filtering for predictable slicer behaviour.
- Data types validated (percentages, currency, whole numbers) to support correct DAX aggregation.
- A dedicated measures table was created to keep all DAX measures organised separately from raw columns.

12. DAX Measures & KPIs

A set of reusable DAX measures power the KPI cards and visuals across all four dashboard pages. Key measures include:

Measure	DAX Logic (simplified)
Total Revenue	SUM(fact_supply_chain[revenue_generated])
Total Products Sold	SUM(fact_supply_chain[units_sold])
Avg Shipping Cost	AVERAGE(fact_supply_chain[shipping_cost])
Avg Defect Rate	AVERAGE(fact_supply_chain[defect_rate])
Total Stock	SUM(fact_supply_chain[stock_level])
Avg Manufacturing Cost	AVERAGE(fact_supply_chain[manufacturing_cost])
Avg Profit Margin	AVERAGE(fact_supply_chain[profit_margin])
Total Logistics Cost	[Total Shipping Cost] + [Total Manufacturing Cost]
Avg Supplier Lead Time	AVERAGE(fact_supply_chain[lead_time_days])

13. Dashboard Development

The report is organised into four pages, each targeting a specific audience and decision area. Every page shares a common slicer panel (Product Type, Supplier Name, Transportation Mode / Inspection Result) so users can drill down consistently across the whole report.

13.1 Executive Overview

The landing page gives leadership a snapshot of overall performance: total revenue, products sold, average shipping cost, and average defect rate, alongside revenue and defect breakdowns by product type and supplier.

- Total Revenue: Rs. 577.60K | Total Products Sold: 46K
- Avg Shipping Cost: Rs. 5.55 | Avg Defect Rate: 2.28%
- Skincare leads revenue at Rs. 0.24M; Supplier 1 leads at Rs. 158K.
- Air freight has the highest average shipping cost (Rs. 6.02) of all four modes.



Figure 13.1 — Executive Overview dashboard page.

13.2 Inventory & Supplier Analysis

This page focuses on inventory health and supplier reliability, combining stock levels, production volumes, defect rates, and manufacturing costs into one supplier scorecard-style view, backed by a detailed drill-through table.

- Total Stock: 5K units | Total Production Volume: 57K units
- Avg Manufacturing Cost: Rs. 47.27 | Avg Defect Rate: 2.28%
- Haircare holds the highest stock level (1,644 units); Supplier 2 leads production volume (14.1K units).
- Supplier 5 shows the highest average defect rate (2.67%), flagging a quality-control priority.

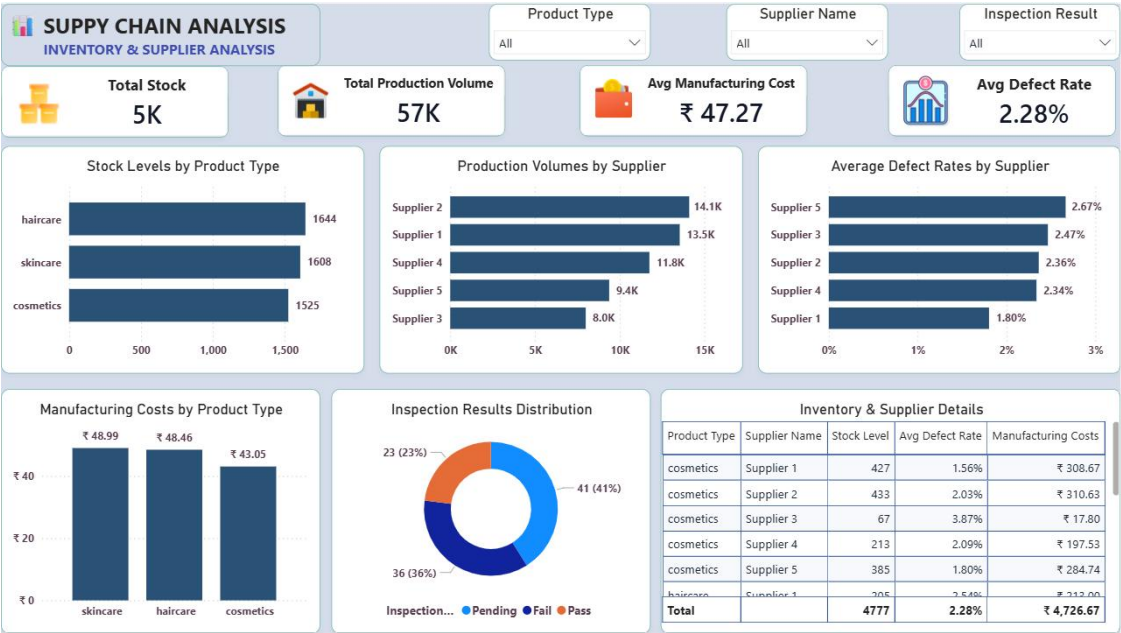


Figure 13.2 — Inventory & Supplier Analysis dashboard page.

13.3 Logistics & Profitability Analysis

This page connects logistics performance to profitability, comparing shipping cost and time across transportation modes and product types, and ranking suppliers by average lead time.

- Total Shipping Cost: Rs. 554.81 | Avg Profit Margin: 98.76%
- Total Shipping Time: 575 days | Total Logistics Cost: Rs. 52.92K
- Cosmetics delivers the highest profit margin (99.16%); Rail carries the highest total shipping time.
- Supplier 3 has the longest average lead time (20.13 days) among all suppliers.



Figure 13.3 — Logistics & Profitability Analysis dashboard page.

13.4 Business Insights

The final page distils the first three pages into plain-language insights and recommendations, grouped by Executive, Inventory & Supplier, and Logistics & Profitability themes, so non-technical stakeholders can act on the findings directly.

- Highest Revenue Product: Skincare | Top Revenue Supplier: Supplier 1
- Highest Stock Category: Haircare | Highest Profit Margin Product: Cosmetics
- Each insight card is paired with a corresponding, actionable recommendation.

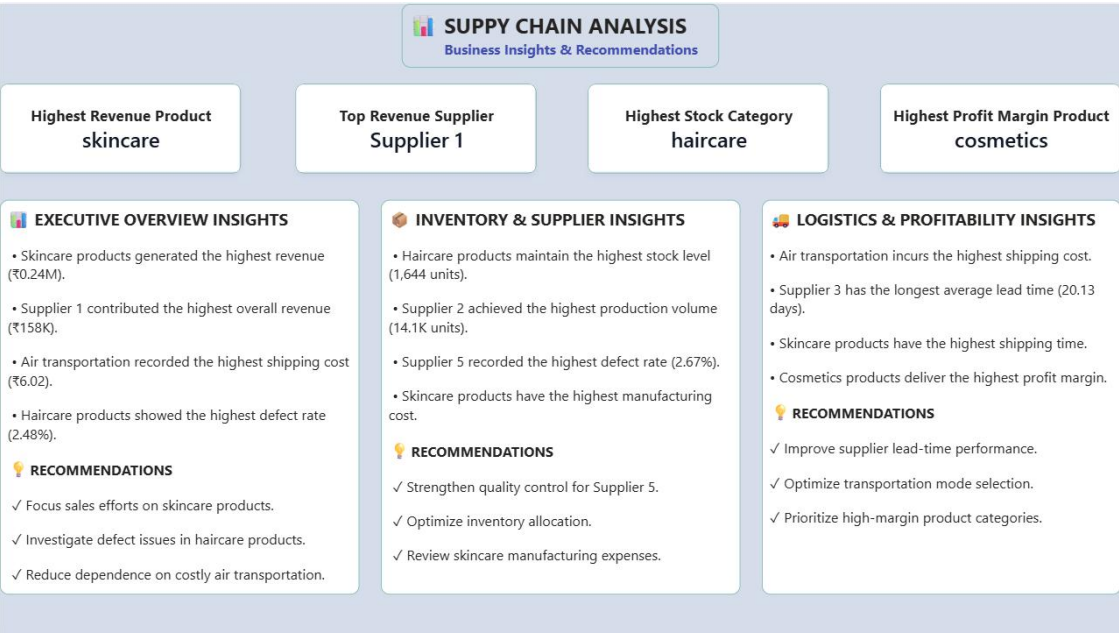


Figure 13.4 — Business Insights dashboard page.

14. Key Insights

- **Skincare** is the strongest-performing category, generating the highest revenue (Rs. 0.24M) and the highest profit margin among top performers.
- **Supplier 1** is the top revenue contributor (Rs. 158K) and also has the lowest average defect rate (1.80%), making it the most reliable supplier overall.
- **Supplier 5** combines the lowest production volume with the highest defect rate (2.67%), marking it as the weakest link in the supply chain.
- **Air transportation** is consistently the most expensive shipping mode (Rs. 6.02 average), while Sea is the cheapest (Rs. 4.97).
- **Haircare** products carry the highest defect rate (2.48%) despite having the largest stock levels, suggesting a quality-cost trade-off worth investigating.
- **Supplier 3** has the longest average lead time (20.13 days), which could create downstream fulfilment risk.
- **Profit margins** stay high across all product types (98.5%–99.2%), indicating pricing is well above landed cost, with cosmetics slightly ahead of the rest.

15. Business Recommendations

Revenue & Product Strategy

- Prioritise marketing and shelf space for skincare, the top revenue and margin driver.
- Investigate the root cause of haircare's elevated defect rate before scaling its production further.

Supplier Management

- Introduce stricter quality checks and corrective action plans for Supplier 5.
- Consider shifting a larger share of production volume toward Supplier 1, given its strong revenue and lowest defect rate.
- Work with Supplier 3 on lead-time reduction, e.g. safety stock or an alternate secondary supplier.

Logistics & Cost Optimisation

- Shift a greater share of shipments from Air to Road or Sea where delivery-time requirements allow, to reduce average shipping cost.
- Reassess routing for skincare, which currently carries the highest shipping times, to protect its strong margin.

16. Project Workflow

1. Collect raw supply chain data (product, supplier, logistics, and quality records).
2. Clean and standardise the data using Power Query and SQL.
3. Design a normalised relational schema and load data into MySQL.
4. Run exploratory SQL analysis to validate early business questions.
5. Import data into Power BI and build the star-schema data model.
6. Create DAX measures for all core KPIs.
7. Design and build the four report pages with consistent slicers and visuals.
8. Derive key insights and translate them into business recommendations.

9. Review, test, and publish the final report.

17. Challenges & Solutions

Challenge	Solution
Inconsistent text values across product and supplier fields	Standardised naming via Power Query transformations before loading
Missing values in defect rate and shipping cost columns	Applied SQL-based validation and imputation rules during ETL
Multiple slicers causing cross-page filter conflicts	Set relationship filter direction explicitly and tested each slicer combination
Keeping DAX measures organised as the model grew	Created a dedicated, disconnected measures table
Fitting dense KPI cards and charts on a single dashboard page	Grouped visuals by theme and used consistent card sizing/spacing

18. Conclusion

This project demonstrates a complete analytics workflow, from raw, messy supply chain data to a polished, decision-ready Power BI report. By combining structured SQL analysis with a well-modelled Power BI report and clear DAX measures, the dashboards surface actionable findings — from which product category drives the most profit, to which supplier and transportation mode need attention. The result is a reusable analytical framework that a supply chain team could apply to monitor performance on an ongoing basis.

19. Future Enhancements

- Connect the model to a live data source for real-time (or near real-time) refresh instead of a static snapshot.
- Add time-series/trend analysis once historical data spanning multiple periods is available.
- Introduce forecasting (e.g. demand or defect-rate forecasting) using Power BI's built-in analytics or an external ML model.
- Add row-level security so each supplier or regional manager sees only their relevant data.
- Build a mobile-optimised report layout for on-the-go monitoring.
- Incorporate cost-of-quality analysis to quantify the financial impact of defect rates by supplier.